

7 Waves

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A progressive wave transfers energy without a net transport of matter. Its speed is $v = f\lambda$. Transverse displacement is perpendicular to propagation; longitudinal displacement is parallel. Intensity scales as amplitude squared for otherwise comparable waves. Superposition adds simultaneous displacements algebraically.

Worked example: follow the reasoning

For $f = 200 \text{ Hz}$ and $\lambda = 1.5 \text{ m}$, wave speed $= 300 \text{ m s}^{-1}$. Doubling amplitude multiplies intensity by four if conditions stay the same.

Quick check · multiple choice

A How is wave speed related to frequency and wavelength?

[1]

A $v = f/\lambda$

B $v = \lambda/f$

C $v = f\lambda$

D $v = f + \lambda$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A wave travels at 340 m s^{-1} and has frequency 680 Hz . Find wavelength and period.

[3]

Final answer: _____

Answers and explanation

Quick check: C. $v=f\lambda$

Question 1 [3 marks]

$\lambda = v/f = 0.50 \text{ m}$ [2]; $T = 1/f = 1.47 \times 10^{-3} \text{ s}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.